

GMAT Study App (On-demand, personalized) — One-page product spec

Summary

A web app that helps users study for the **GMAT Focus** (Quant, Verbal, Data Insights) by running a **baseline mini-exam**, estimating strengths/weaknesses by topic, and generating an adaptive **study plan** with **guided practice**. Questions are generated **on-demand per user session** from structured specs and validated for correctness and quality.

Target user + goal

- **User:** GMAT test-taker who wants a clear plan and high-quality practice.
- **Goal:** Improve score efficiently via personalized practice and review.

Core user flows

- **Onboarding**
 - Select test date (or “not set”), availability (mins/day), target score (optional)
 - Start baseline mini-exam
- **Baseline mini-exam → Diagnostics**
 - 30-question diagnostic (10 Quant, 10 Verbal, 10 Data Insights)
 - Results: section performance, top weak topics, speed vs accuracy flags
- **Study plan**
 - Weekly schedule + “Today’s session” queue (20–60 min)
 - Mix of targeted drills, mixed review, timed sets, and redo queue
- **Guided practice**
 - Attempt question → optional hint ladder → submit → review
 - “Tested concept” is **always present but hidden** (user can unhide)
 - Errors tagged and scheduled for spaced review
- **Checkpoint loop**
 - Every 1–2 weeks: short reassessment quiz → plan updates

Question generation (no admin UI)

- **On-demand generation** driven by `QuestionSpec`:
 - Inputs: user skill estimates, plan needs, desired section/type/topic/difficulty
- **Generators**
 - Quant: parameterized templates (provable single correct answer)
 - Verbal: structured prompt generation + consistency checks

- Data Insights: synthetic dataset/table/graph + question(s)
- **Validators (must-pass)**
 - correctness (Quant/DI recomputed)
 - single-correct-option
 - style/clarity rules + time sanity
 - dedupe vs recent user history
- Persist each served question for review and reproducibility.

Baseline exam blueprint (v1)

- **Quant (10)**: arithmetic/algebra core mix + 2 word problems + 2 stats/counting rotation
- **Verbal (10)**: 6 CR + 1 RC passage with 4 questions
- **Data Insights (10)**: graph/table (4) + two-part (3) + multi-source set (3)

Personalization logic (v1)

Maintain mastery + uncertainty per topic. Next-question selection optimizes:

Subject to:

- interleaving constraints
- periodic mixed review
- limited introduction of brand-new subtopics

Key screens (MVP)

- Landing / Sign-in
- Onboarding
- Baseline Exam (with timer + progress)
- Results Dashboard (section + topic breakdown)
- Study Plan (weekly + today queue)
- Practice Player (question, hint ladder, reveal tested concept, solution review)
- Review / Redo Queue (missed/guessed/timeouts)

Data model (minimum)

- User
- Topic (taxonomy tree)
- GeneratedQuestion (spec, seed, rendered content, solution, testedConcept, generatorVersion, validationReport)
- Attempt (answer, correct, time, hintsUsed, testedConceptShown, confidence, errorType)
- SkillEstimate (userId, topicId, mastery, uncertainty, updatedAt)

- StudyPlan + PlanItem (scheduled tasks)

Success metrics

- Baseline completion rate
- Weekly active days
- Accuracy + time improvements per topic
- Reduction in repeat error types
- Retention (week 1 → week 4)

Open decisions (next)

- Support full-length simulated exams in v1 or v2?
- Timing strictness in practice mode (off / soft / hard)?
- Initial Quant template set (coverage vs speed-to-ship)?